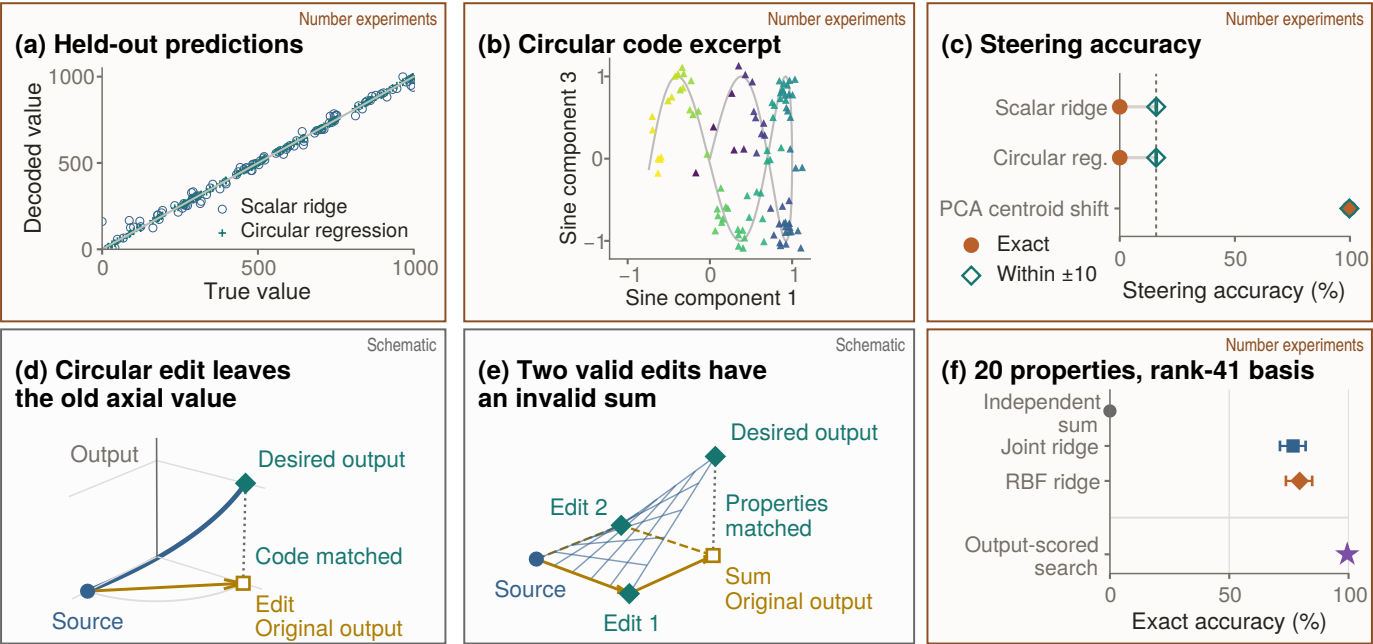

Number experiments
(a) Held-out predictions
Decoded value
1000
500
0
0
500
1000
True value
Scalar ridge
Circular regression
Number experiments
(b) Circular code excerpt
Sine component 3
1
0
−1
−1
0
1
Sine component 1
Number experiments
(c) Steering accuracy
Scalar ridge
Circular reg.
PCA centroid shift
Exact
Within ±10
0
50
100
Steering accuracy (%)
Schematic
(d) Circular edit leaves the old axial value
Output
Desired output
Code matched
Edit
Original output
Source
Schematic
(e) Two valid edits have an invalid sum
Desired output
Edit 2
Properties matched
Source
Sum
Original output
Edit 1
Number experiments
(f) 20 properties, rank-41 basis
Independent sum
Joint ridge
RBF ridge
Output-scored search
0
50
100
Exact accuracy (%)